

Mean |SHAP| of temperature at each lag (0, 1, 3, 7, 14, 21, 30 d) from interventional TreeSHAP, 50 bootstrap replicates.
Ribbon = 95% percentile CI. Marker fill encodes directional asymmetry (pct_positive: fraction of patient-visits with SHAP > 0).



Organ system

- Cardiovascular
- Renal
- Haematological
- Immunological
- Lipid
- Body composition

Markers

- temp ↑ biomarker (>55%)
- mixed direction (45-55%)
- temp ↓ biomarker (<45%)

Takeaways

- 4 of 10 biomarkers peak within 0-3 d; 7 peak at either edge of the window (0-3 d or 21-30 d).
- Cardiovascular markers show the most symmetrical responses with SHAP balanced across positive and negative directions.
- Haematological markers (Hct, Hb) peak at lag 1/1 d, consistent with acute haemoconcentration within days of hot weather.
- Body-composition markers peak at lag 7/14 d (medium-horizon exposure).
- CI ribbons: 50 bootstrap replicates, 95% percentile interval.